

Generation Interconnection System Impact Study Report

For Queue Project Q18-0912

Mid America 138 kV

January 19, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0912. The Interconnection Customer proposes a 96.8 MW generating facility interconnecting at the Mid America 138 kV substation in the Pennsylvania area. The request was received on January 04, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 254 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$1,951,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
MOUNTAIN 500/345 kV Transformer Bank	LITTLE CHUTE SUBSTATION-FT HUMBURG 345.0 kV	106.8	1,195	5.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MTN 500 kV/345 kV XFMR	Replace transformer with higher-rated unit	\$28,393,000	7% (\$1,951,000)
	Total allocated to project		\$1,951,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.